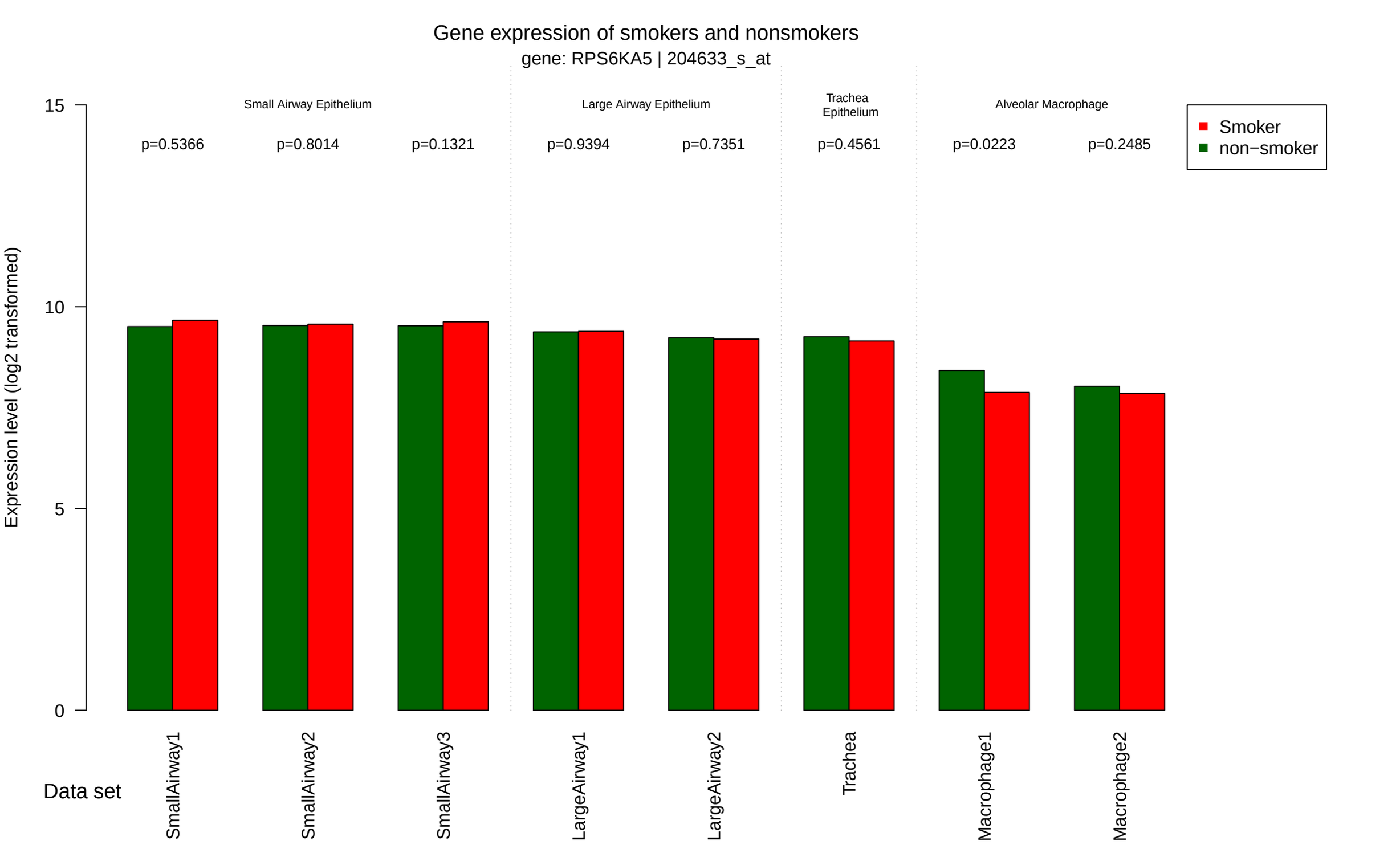
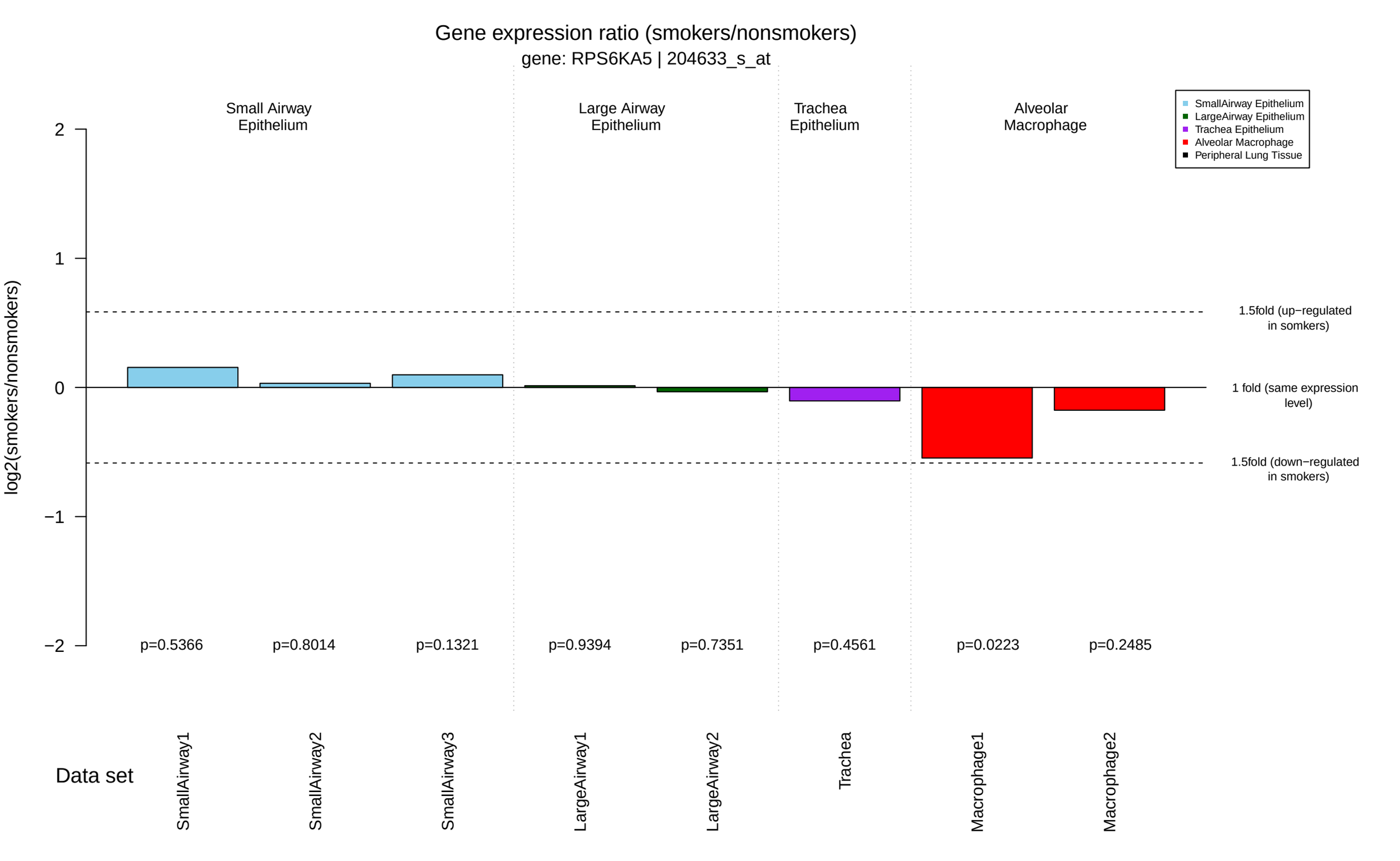
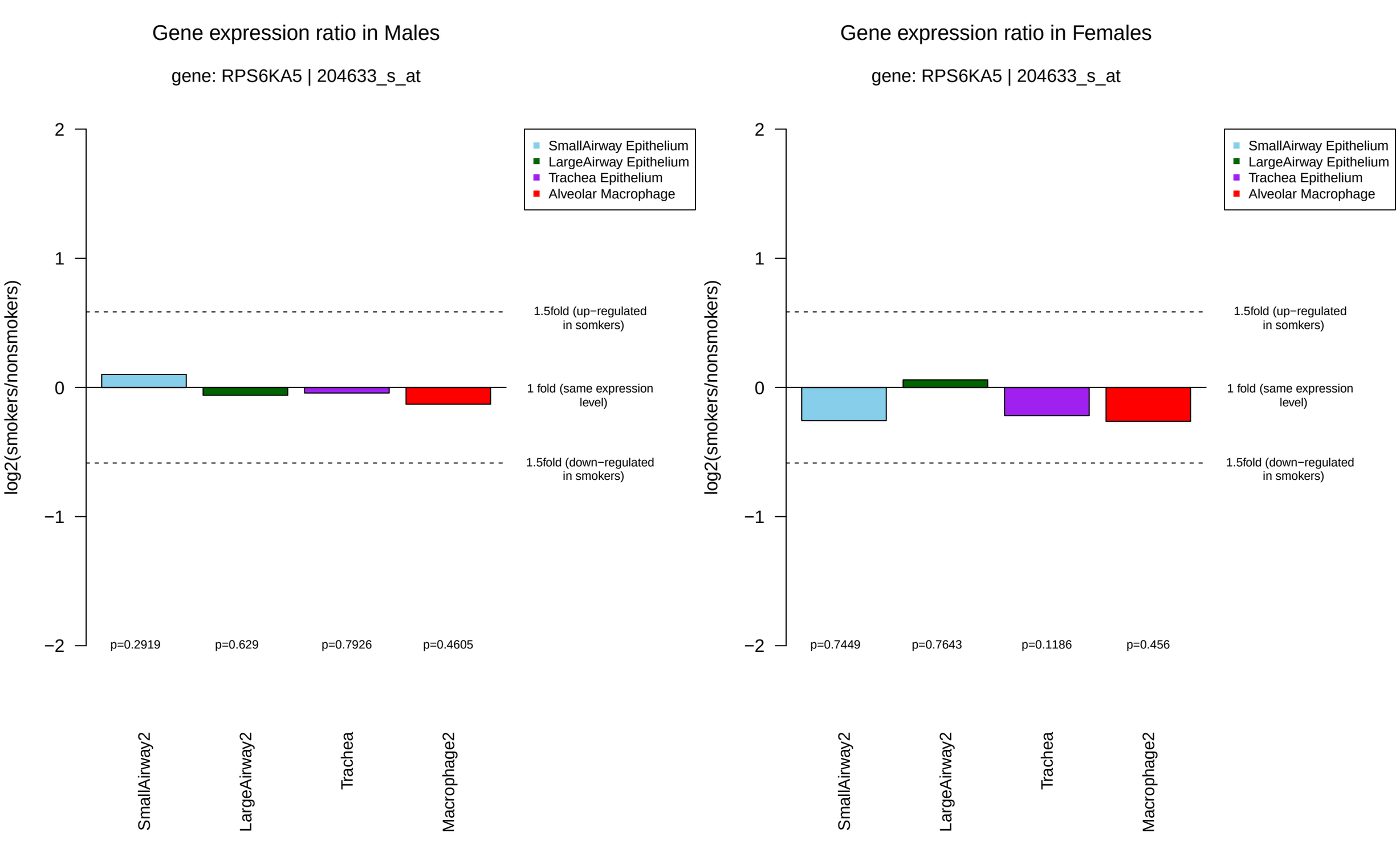
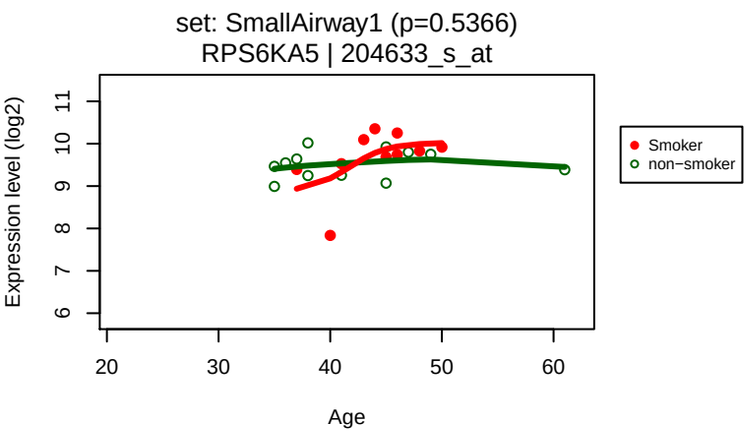
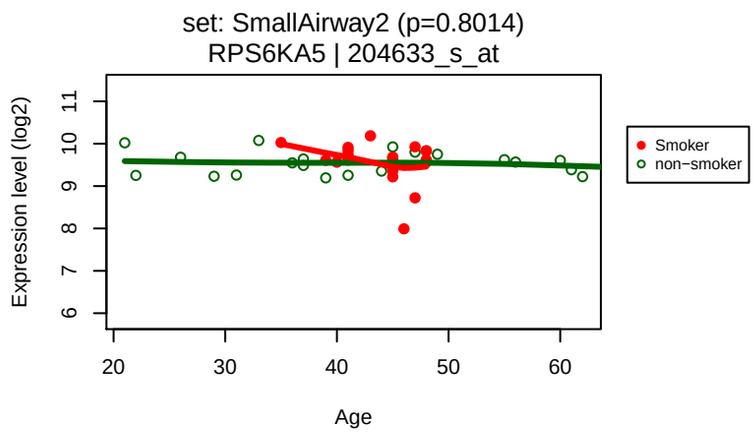
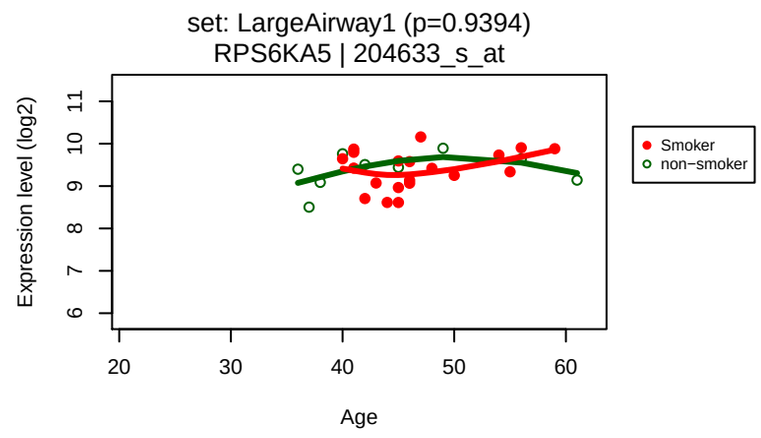

Gene expression of smokers and nonsmokers
gene: RPS6KA5 | 204633_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5366
p=0.8014
p=0.1321
p=0.9394
p=0.7351
p=0.4561
p=0.0223
p=0.2485
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

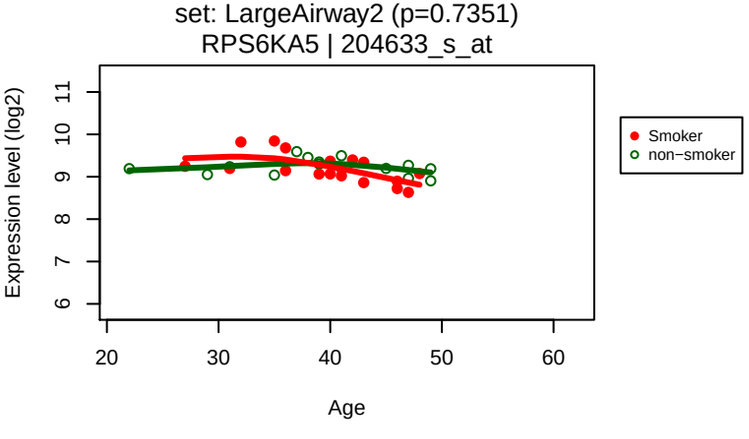

Gene expression ratio (smokers/nonsmokers)
gene: RPS6KA5 | 204633_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5366
p=0.8014
p=0.1321
p=0.9394
p=0.7351
p=0.4561
p=0.0223
p=0.2485
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

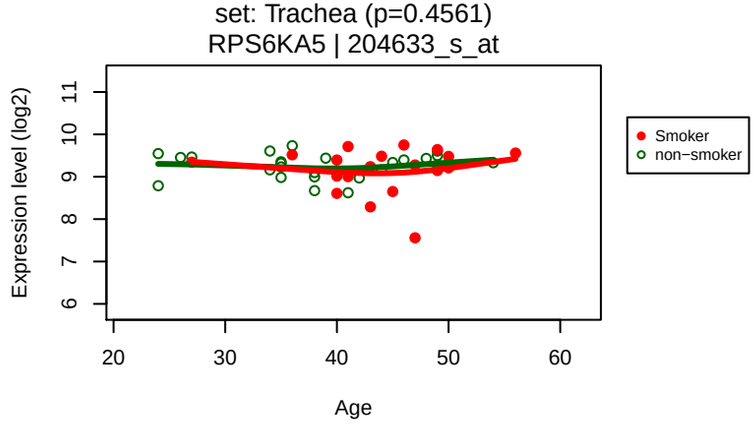

Gene expression ratio in Males
gene: RPS6KA5 | 204633_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2919
p=0.629
p=0.7926
p=0.4605
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPS6KA5 | 204633_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7449
p=0.7643
p=0.1186
p=0.456
SmallAirway2
LargeAirway2
Trachea
Macrophage2

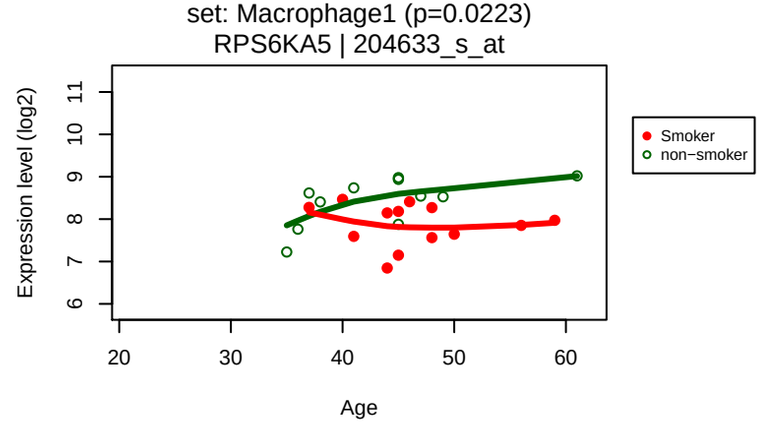

set: SmallAirway1 (p=0.5366)
RPS6KA5 | 204633_s_at
Expression level (log2)
Age
Smoker
non−smoker

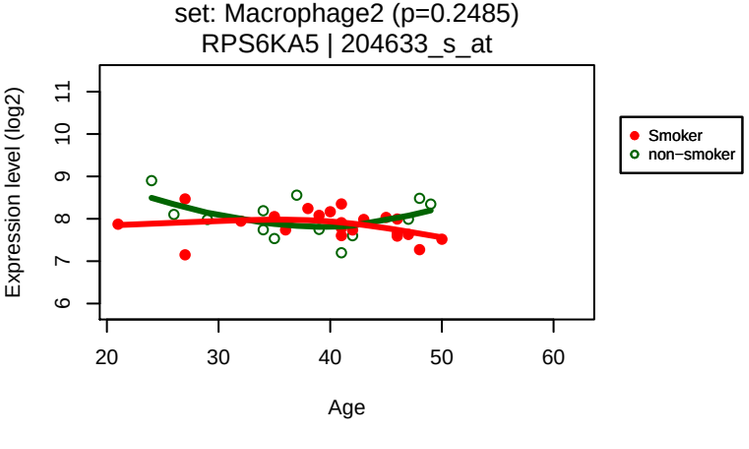

set: SmallAirway2 (p=0.8014)
RPS6KA5 | 204633_s_at
Expression level (log2)
Age
Smoker
non−smoker

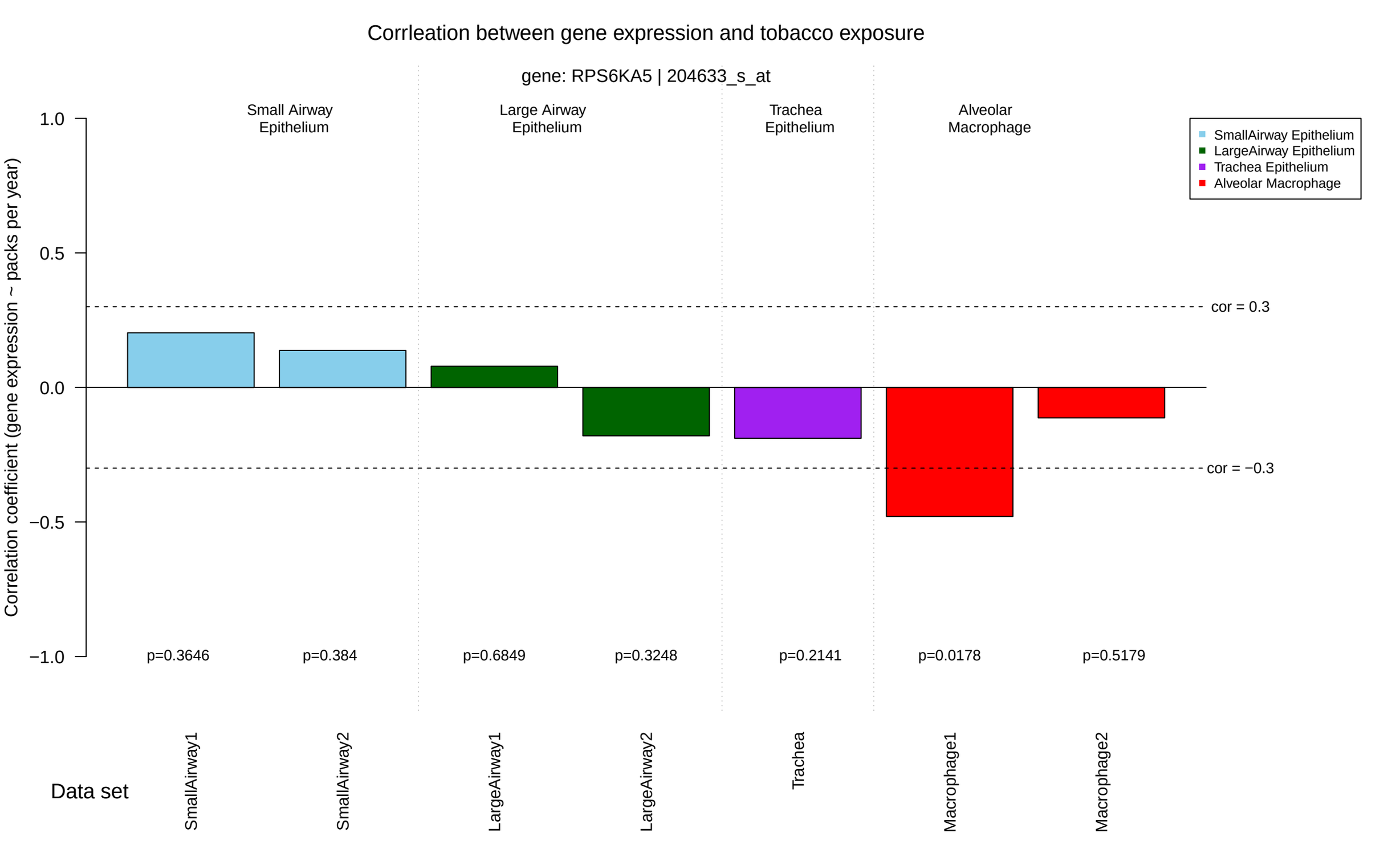

set: LargeAirway1 (p=0.9394)
RPS6KA5 | 204633_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7351)
RPS6KA5 | 204633_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4561)
RPS6KA5 | 204633_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0223)
RPS6KA5 | 204633_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2485)
RPS6KA5 | 204633_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPS6KA5 | 204633_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3646
p=0.384
p=0.6849
p=0.3248
p=0.2141
p=0.0178
p=0.5179
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2